

Steady State Cost in Competing Aggregation-Fragmentation Dynamics

Fitzwilliam James Keenan-Koch^{1,*}

¹Vermont Complex Systems Institute, University of Vermont, Burlington, VT, USA

*fkeenank@uvm.edu

ABSTRACT

Aggregation and fragmentation among polymers has long been an area of active research and utility in physical chemistry. Drawing inspiration from the dynamics of social formulation, we make a preliminary effort at minimizing cost and probability of domination among various potential fragmentation kernels. This work has relevance for any work formalizing the formulation, dissolving, and regulation of groups of abstract individuals. Social groups are of particular interest, from musical subcultures to the regulation of corporate consolidation.

1 Introduction

Things fall apart, Yeats wrote¹. Of course, they also come together. What balance is struck between the two, and crucially, **what does this balance cost?**

To attempt an answer to this question, we turn to the physics of aggregation and fragmentation, the competing processes by which particles combine to form larger clusters and break up into smaller ones. As thoroughly explicated in *A Kinetic View of Statistical Physics*, aggregation and fragmentation can be thought of as an infinite set of differential equations over all cluster sizes from one to infinity². Rates of change are based on current amounts of clusters and a kernel specifying reaction rates between clusters of various sizes. Clusters of size one are termed monomers, and the formulation as a whole is termed the Smoluchowski equations.

While Krapivsky *et al* term such differential equations master equations, their equations make a mean-field assumption: they track the *expected* cluster size distribution and assume it deterministic rather than keeping track of the full spread of probability over the stochastic evolution of the system. In other words, they track the *count* of the cluster sizes, rather than the probability of *each count* of each cluster.

A more complete accounting is given by the Marcus–Lushnikov process, in which $ML^{(n)}(x, t)$ is a random number whose probability evolves according to a Markov Chain involving all other $ML^{(n_i)}(x, t)$ ³. These models typically rescale the total mass of the system to be 1, and are identical to the Smoluchowski equations if the kernel is homogenous, or as M , the number of initial unitary particles considered, goes to infinity. However, these models are far less tractable, and even in 2018 computing the time-evolving probability with only 48 initial monomers took around 3000 minutes⁴.

Regardless of the form, prior work has largely focused on either the time-evolution of cluster sizes or the static steady state distribution of them. Instead, this work focus on the interplay between the steady state rate of fragmentation, or the *cost* of maintaining the steady state, and the distribution of cluster sizes. More simply, using the Smoluchowski equation form, this work charts the pareto front between minimizing the steady state probability of having an oversized *dominant* cluster and minimizing the number of units of mass continually split by fragmentation.

2 Methods:

The problem

We model competing aggregation and fragmentation over a variety of rate kernels in order to study the steady-state distribution of cluster sizes as well as the steady-state rate of fragmentation measured in units of mass removed from larger clusters. Our question is simple: given a pattern of aggregation, what fragmentation technique minimize the proportion of dominant clusters and steady-state fragmentation rate? Does this dual-minimization create a pareto front?

Preliminaries

Aggregation

Aggregation is generally the transition

$$c_i + c_j \xrightarrow{K_{ij}} c_{i+j}$$

where:

- M is the total amount of mass divided by the size of the smallest cluster (which is 1).
- c is a cluster
- i and j are masses, or sizes
- K_{ij} is the rate of aggregation between clusters of mass i and mass j into a clusters of mass $i + j$
- K , then, is a symmetric matrix $K_{ij} = K_{ji}$ of dimensions $M \times M$
- The clusters $\vec{c}$ are all well-mixed and come into ‘aggregation radius’ with a rate determined solely by their quantity.

Then the continuous ordinary differential equation for all cluster sizes c_k is the sum of aggregation rates producing a cluster size and the rate of aggregation reducing the amount of a cluster size, with each weighted by the current amount of the relevant clusters:

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{i \geq 1} K_{ik} c_i \quad (1)$$

$$= \frac{1}{2} \sum_{i,j \leq 1} K_{ij} c_i c_j [\delta_{i+j,k} - \delta_{i,k} - \delta_{j,k}] \quad (2)$$

where δ is the Kronecker delta.

Fragmentation

Fragmentation is generally the transition

$$c_{i+j} \xrightarrow{F_{ij}} c_i + c_j$$

where F is a symmetric matrix of size $M \times M$ and F_{ij} is the rate of fragmentation of a cluster of size $i + j$ into a cluster of size i and another of size j . The ordinary differential equation for fragmentation across all clusters is then:

$$\frac{dc_k}{dt} = \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} F_{ij} \quad (3)$$

Aggregation and Fragmentation

Putting equations (1) and (3) together and scaling them by parameterized rates α and β yields the main differential equation we use in our model:

$$\frac{dc_k}{dt} = \dot{c}_k = \frac{1}{2} \sum_{i+j=k} \alpha K_{ij} c_i c_j - c_k \sum_{j \geq 1} \alpha K_{kj} c_j + \sum_{j \geq 1} \beta F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} \beta F_{ij} \quad (4)$$

The Model

Using equation (4) as our basis, we let $T = 100$ be the total time the integrator runs, and $M = 100$ be the total amount of mass, which is conserved across all steps. Note that in calculations we use a normalized representation of mass $N = 1$, so the initial choice of M affects only the speed of our calculation and the resolution of our results; in other words, M is effectually the number of discrete ‘buckets’ of cluster sizes. Higher M produces slower, but more granular, integrations. With this formalization in place, we use the `scipy` Python package to integrate the differential equation and thus calculate the time evolution of the system. From this time evolution we are able to easily calculate the earliest time index s at which the state of the system ceases to change and thus reaches a steady state.

Cost

The main emphasis of this work is to keep track of the time evolution not only of the cluster sizes, but of the *rate of fragmentation*. More specifically, we are interested in the rate at which unit masses are broken off of clusters; the number of unit masses that break at each infinitesimal dt . Letting κ indicate this cost, we can write its rate as:

$$\dot{\kappa} = \sum_{k=1}^M \sum_{i+j=k} \beta F_{ij} c_k(t) \times \min(i, j) \quad (5)$$

where we write only one fragmentation sum because the sum of mass leaving all cluster sizes double counts with the sum of mass entering all cluster sizes. To calculate the full time-evolution of cost, we augment the state vector with an additional variable c_κ with associated differential equation (5).

Therefore when the integrator finishes running we have $\vec{\kappa}$, the vector of how κ changes over the full time-evolution of the system. However, what we are principally interested in is $\dot{\kappa}$, the continuous rate; more particularly, we are interested in $\dot{\kappa}_s$, the rate of fragmentation *when the system is in a steady state*. To derive this, we simply calculate the gradient of $\vec{\kappa}$ from $t = s$ onwards using `numpy`.

Domination

The goal of this model is to simulate certain social systems just as much as it is physical. As such, we are interested not in gelation, when all mass in the system converts to a single cluster, but in **Domination**, when a cluster is larger than the sum of all other clusters. Clearly, this must happen whenever a cluster contains more than 50% of the mass. Generally, however, we are able to say that the proportion of mass in clusters greater than some dominating threshold x is the weighted ratio:

$$g_x(t) = \frac{\sum_{k>xM} c_k(t)k}{\sum_1^M c_k(t)k} \quad (6)$$

Where the denominator is always 1 but is left included for completeness and to account for any numerical drift. To calculate some g_x in practice, we take the distribution of clusters at the steady state $t = s$ and complete equation (6); we are interested in this value only at the steady state of the system and leave this implicit in the notation. Throughout, we report generally on $g_{0.6}$, the proportion of mass in clusters greater than $.6M$, though $g_{0.8}$ is calculated as well.

Kernels

Recall that the only requirement for our kernels K and F is that they are symmetric. It is then easy to see that there is an infinite set of kernels, even without our scaling rates α and β ; consider, for example, $(i+j)^p \forall p \in \mathbb{R}$. For this reason, we use kernels drawn from a combination of physics literature and inspiration from social phenomena.

First, aggregation. Consider our clusters as polymer chains consisting of k monomers, each with all monomers having the same amount of f -functional reactive endgroups. Clearly, anytime a monomer joins a polymer chain, two endgroups are consumed and f are added, as in equation (7). If all endgroups are equally reactive, then the reaction rate between two clusters of sizes i, j will equal the product of the number of endgroups. Together, we can write this formulation as follows:

$$kf - 2(k-1) = (f-2)(k+2) \quad (7)$$

$$g(i, j) = (f-2)^2 ij + 2(f-2)(i+j) + 4 \quad (8)$$

When $f = 2$, clearly equation (8) yields a constant kernel, when $f \rightarrow \infty$, it yields a product kernel, and anywhere in between it is a linear combination of a constant, sum, and product kernel. In social phenomena, we can interpret the number f to indicate *how many social bonds* a person is able to maintain within the social context. In 1992, Dunbar used neocortex size and extrapolation from social dynamics observed in other primates to argue that humans can maintain 150 stable relationships⁵. We use this number to construct the **dunbar** kernel. We might also argue that in the context of a large corporation, for example, the number of functional ‘connections’ *homo corporatum* might have with another is practically infinite; we use this argument to construct the **product** kernel. In practice, the two are functionally identical. For good measure, we throw in the **sum** and **constant** kernels as well. Clearly, moving from the constant to sum to product kernels will accelerate the rate of aggregation, all other things being equal.

Next, fragmentation, again drawing inspiration from both prior physics and social phenomena. For example, the **chipping1** and **chipping5** models do exactly what they sounds like: chip away blocks of either one or five masses only. The **rprod** model, on the other hand, is something like ‘too big to fail’: large clusters are unlikely to break apart.

Our logic and equations See Table 1 and explain how we arrived at dunbar, polymerization, etc.

Name	Equation
Chipping1	$F_{ij} = \beta(\delta_{i,1} + \delta_{j,1})$
Chipping5	$F_{ij} = \beta(\delta_{i,5} + \delta_{j,5})$
Sum	$F_{ij} = \beta(i + j)$
Product	$F_{ij} = \beta(ij)$
Constant	$F_{ij} = \beta$
Reverse Product	$F_{ij} = \frac{\beta}{ij}$
Dunbar	$F_{ij} = \beta\left(ij + \frac{2(i+j)}{150} + \frac{4}{150^2}\right)$

Table 1. Specification of the seven kernels used for aggregation and fragmentation. Note that all kernels are also used for aggregation by substituting α for β .

Third-level section

3 Results

The results can contain subheadings as desired and can discuss the results directly.

4 Discussion

The Discussion should be succinct. It typically involve any big picture discussions not covered in Sec. 3, i.e., the Results section. This was an example of a section reference using the `label` and `ref` commands. Figures and tables can also be referenced in LaTeX using the `label` and `ref` commands, e.g. Fig. 1 and Tab. 3.

This discussion can therefore mention scientific consequences of the results, ideas for future work, stress approximations or simplifications that limit the applicability of the model, or discuss things you tried that did not work. The Discussion section must not contain subheadings.

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Additional information

Data availability statement: No real world data is used in this project

Code availability statement: Code is available at github.com.

References

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Figure 1. Legend (350 words max). Example legend text for an example figure. Figures should be in pdf format unless it is a photo or archival image not in vector graphics, as is the case here (in which case jpg and png are acceptable).

Table 2. Model Card

1. Model Details

The problem
Modeling approach
Degrees of freedom
Number of parameters
Implementation language
Relevant publications
License
Contact information

2. Intended Use

Primary intended uses
If forecasting, validation metrics
If applicable, data incorporated
Primary users
Out-of-scope uses

3. Key assumptions

Population size
Contact structure
Time structure and memory effects
Other

4. Metrics

Computational complexity
Variation approaches

8. Ethical Considerations

Potential risks and harms
Bias and fairness concerns
Mitigation strategies
Impact on stakeholders

9. Suggested next steps

Issues with implementations
Recommendations for use
Future work

Condition	n	p
A	5	0.1
B	10	0.01

Table 3. Legend (350 words max). Example legend text for an example table.